

Machine Learning Data Annotation (MLDA)

Customer-Service Intent Classification

Operations Report

Synthetic end-to-end annotation, quality assurance, compliance, and ML evaluation project

Project item	Result
Synthetic conversations	5,000
Intent categories	20
Annotation setup	3 simulated annotators + adjudication
Validation accuracy	98.20%
Final test accuracy	99.40%
Test errors	3 / 500
PII pattern checks	No detected email, phone, URL, or card-like values

1. Executive Summary

This project simulates a production-style Machine Learning Data Annotation workflow for an e-commerce customer-service intent classifier. The process covers raw customer messages, intent taxonomy design, annotation guidelines, annotation QA, compliance auditing, ML dataset preparation, model training, validation, testing, and error analysis.

The final classifier uses TF-IDF text features with Logistic Regression. It achieved 98.20% validation accuracy and 99.40% test accuracy on the synthetic dataset. Three test errors were reviewed individually.

Important limitation: all conversations and annotation outcomes are synthetic/simulated. These scores demonstrate the workflow and should not be treated as production performance.

2. Project Objective

The objective is to classify each customer-service message into one primary intent while demonstrating the operational controls required around annotation quality, compliance, and model evaluation. The intended business use case is automatic ticket routing and intent detection.

3. Dataset and Intent Taxonomy

The raw dataset contains 5,000 synthetic customer-service conversations with conversation ID, channel, customer message, language, and synthetic-data metadata. The taxonomy contains 20 intents: order_status, delivery_delay, delivery_estimate, delivery_tracking, return_request, refund_status, payment_failure, duplicate_charge, order_cancellation, damaged_item, wrong_item, missing_item, account_access, password_reset, account_security, address_change, subscription_cancel, subscription_billing, promo_code, and product_information.

4. Annotation Workflow

Formal annotation guidelines define a single-label policy, intent definitions, inclusion/exclusion criteria, multi-intent handling, high-confusion pairs, QA standards, and compliance guidance. The simulated labeled dataset contains three annotator labels, an adjudicated intent, confidence, review status, and reviewer metadata. The gold key is kept separate from the production-style labeled dataset.

5. Annotation Quality Assurance

QA used annotator-to-adjudicated-label agreement, pairwise Cohen's kappa, and an annotation confusion matrix. Recurring boundary areas included return_request vs refund_status, payment_failure vs subscription_billing, duplicate_charge vs subscription_billing, delivery_estimate vs delivery_tracking, and wrong_item vs damaged_item. These findings show why annotation guidelines should be refined when recurring ambiguity is discovered.

6. Compliance Audit

A compliance script checks required fields, duplicate conversation IDs, and common PII patterns including email addresses, phone numbers, URLs, and card-like numeric sequences. The dataset is synthetic and contains no known real customer records. Required production fields are checked separately from optional review metadata so legitimate blanks are not incorrectly treated as missing core data.

7. ML Dataset Preparation

For ML, customer_message is used as the text feature and adjudicated_intent is used as the target. Annotator-specific labels are not used as model features. The data is split into 80% training, 10% validation, and 10% test sets using stratified sampling.

8. Model Training

The model is a scikit-learn Pipeline containing TfidfVectorizer and LogisticRegression. TF-IDF converts text into numerical features, while unigrams and bigrams capture individual words and short phrases. Logistic Regression learns the relationship between those features and the 20 intent classes. The trained pipeline is saved as intent_classifier.pkl.

9. Model Validation and Testing

Validation accuracy was 98.20% on 500 conversations, with 9 incorrect predictions. Final held-out test accuracy was 99.40% on 500 conversations, with 3 incorrect predictions. The test result corresponds to 497 correct predictions out of 500. Because the dataset is synthetic and relatively clean, these scores primarily demonstrate successful learning on the constructed dataset.

10. Model Error Analysis

Three test errors were identified. CONV-01647 was labeled missing_item but predicted damaged_item; the message explicitly mentions a damaged item, suggesting a possible label inconsistency. CONV-01485 was labeled return_request but predicted refund_status; the message contains both concepts and represents a primary-intent boundary case. CONV-03815 was labeled duplicate_charge but predicted payment_failure; the message describes a rejected payment, suggesting the stored label may be questionable.

11. Key Findings

The end-to-end MLDA workflow was successfully implemented. Clear annotation rules are especially important for closely related intents. Annotator disagreement can reveal weak taxonomy boundaries. Compliance checks should focus on required fields and privacy risks. Model errors should be reviewed alongside the labels rather than automatically blamed on the model.

12. Recommended Next Improvements

A stronger production simulation would add a manually reviewed gold sample, richer multi-turn conversations, intentionally ambiguous examples, realistic class imbalance, and a separate unseen evaluation set. Guideline versions

should be updated whenever QA identifies recurring ambiguity.

13. Conclusion

This project demonstrates that customer-service intent classification is not only a modeling task. Taxonomy design, annotation standards, QA, compliance, dataset preparation, model training, evaluation, and error analysis form one traceable MLDA workflow. The model performed strongly on the synthetic dataset, while error analysis showed how improving label quality can be as important as improving the classifier.